

# LABORATORY RECORD

|                |                            |
|----------------|----------------------------|
| Course         | Full Stack Web Development |
| Course Code    | 23CM4121                   |
| Experiment No. | 2                          |
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## 1. TITLE

A24126552096

Design and Implementation of a Styled Webpage Using HTML5 and CSS3

## 2. AIM

To design and implement an attractive static webpage using HTML5 and CSS3 with navigation links, content cards, gradients, text shadows, hover effects, flexbox layout, borders, rounded corners, buttons, and a footer.

## 3. OBJECTIVE

1. To create a structured webpage using HTML5 elements.
2. To link an external CSS file with an HTML document.
3. To apply CSS properties such as background, color, padding, margin, and font styles.
4. To arrange webpage cards using CSS Flexbox.
5. To implement hover effects for navigation links and buttons.
6. To create visually attractive cards using borders, border-radius, and box-shadow.

## 4. THEORY

CSS3 (Cascading Style Sheets) is used to control the presentation and layout of HTML elements. CSS can modify colors, fonts, spacing, borders, shadows, and alignment. Flexbox provides an efficient way to arrange elements horizontally or vertically. Pseudo-class selectors such as :hover allow styles to change when the user interacts with an element. External stylesheets keep webpage structure and presentation separate.

**Application Flow:** HTML5 Structure → External CSS → Navigation Styling → Flexbox Cards → Hover Effects → Footer

## 5. ALGORITHM / PROCEDURE

1. Create an HTML5 document and define the page title as 'Technology Hub'.
2. Link the external stylesheet style.css inside the head section.
3. Create a header containing the main webpage heading.
4. Create a navigation bar with Dashboard, Projects, Courses, and Support links.
5. Create three content boxes for Python, JavaScript, and Database topics.
6. Add headings, descriptions, and Explore buttons inside each box.
7. Create a footer displaying the copyright information.
8. In CSS, style the body, header, navigation, links, content area, boxes, buttons, and footer.
9. Use Flexbox to align the content boxes and add hover effects to links and buttons.
10. Open the HTML file in a browser and verify the layout and styling.

## 6. PROGRAM

### index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Technology Hub</title>
<link rel="stylesheet" href="style.css">
</head>

<body>

<header>
<h1>Technology Hub</h1>
</header>

<nav>
<a href="#">Dashboard</a>
<a href="#">Projects</a>
<a href="#">Courses</a>
<a href="#">Support</a>
</nav>

<section class="content">

<div class="box">
<h2>Python</h2>
<p>Python is a popular programming language used for software development.</p>
<button>Explore</button>
</div>

<div class="box">
<h2>JavaScript</h2>
<p>JavaScript adds functionality and interactive features to websites.</p>
<button>Explore</button>
</div>

<div class="box">
<h2>Database</h2>
<p>Databases are used to store, organize, and manage application data.</p>
<button>Explore</button>
</div>

</section>

<footer>
<p>© 2026 Technology Hub</p>
</footer>

</body>
</html>
```

### style.css

```
body {
margin: 0;
font-family: Arial, sans-serif;
background: #eef5f9;
}

header {
background: linear-gradient(to right, #ff6600, #ff0066);
color: white;
text-align: center;
padding: 25px;
text-shadow: 2px 2px 5px gray;
}

nav {
background: #222;
text-align: center;
padding: 15px;
}

nav a {
color: white;
```

```
text-decoration: none;
margin: 20px;
font-size: 18px;
}

nav a:hover {
color: orange;
}

.content {
display: flex;
justify-content: center;
gap: 25px;
margin: 35px;
}

.box {
background: white;
width: 250px;
padding: 20px;
border: 2px solid #555;
border-radius: 10px;
text-align: center;
box-shadow: 5px 5px 10px #999;
}

button {
background: #0066cc;
color: white;
border: none;
padding: 10px 18px;
border-radius: 5px;
cursor: pointer;
}

button:hover {
background: #ff6600;
}

footer {
background: #222;
color: white;
text-align: center;
padding: 15px;
margin-top: 20px;
}
```

## 7. CODE EXPLANATION

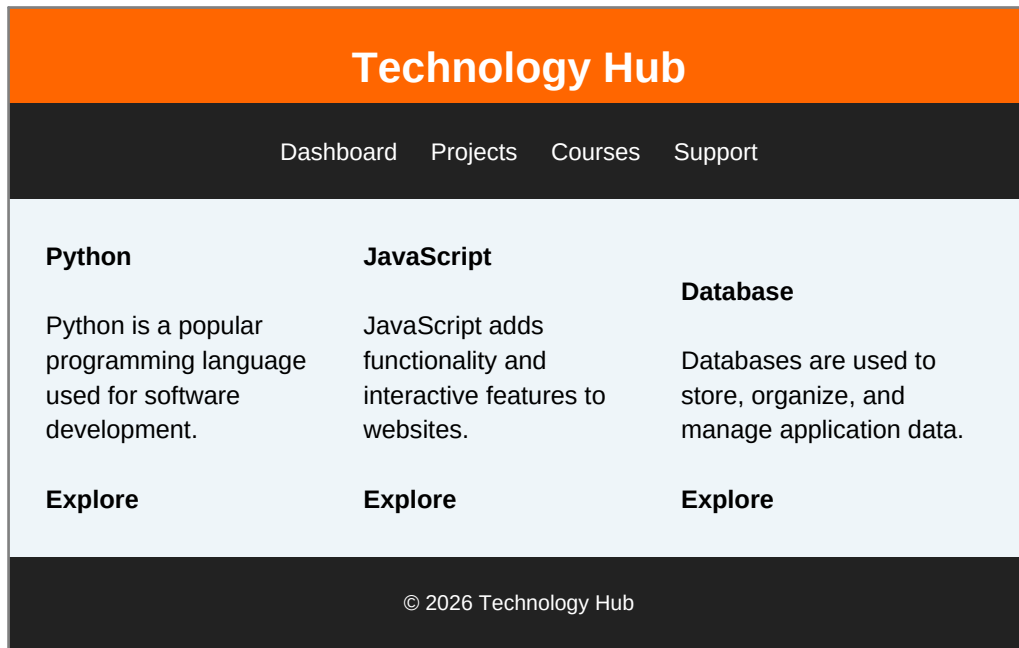
| Code / Syntax               | Explanation                                                                 |
|-----------------------------|-----------------------------------------------------------------------------|
| <header>                    | Defines the header section containing the main webpage heading.             |
| <nav>                       | Creates the navigation section containing webpage links.                    |
| .content { display: flex; } | Uses Flexbox to arrange the content boxes in a row.                         |
| .box                        | Styles each content card with padding, border, rounded corners, and shadow. |
| :hover                      | Changes the appearance of links and buttons when the mouse moves over them. |
| linear-gradient()           | Creates a smooth color transition in the header background.                 |
| box-shadow                  | Adds a shadow effect around the content boxes.                              |

## 8. TEST CASES AND EXECUTION DATA

| Test Case | Input / Action         | Expected Result                     | Actual Result               | Status |
|-----------|------------------------|-------------------------------------|-----------------------------|--------|
| TC-01     | Open webpage           | Header and navigation are displayed | Displayed correctly         | Pass   |
| TC-02     | Inspect cards          | Three topic cards appear in a row   | Cards aligned using Flexbox | Pass   |
| TC-03     | Move mouse over link   | Link color changes                  | Hover effect works          | Pass   |
| TC-04     | Move mouse over button | Button background changes           | Hover effect works          | Pass   |
| TC-05     | Inspect card styling   | Border, radius and shadow appear    | Styling displayed correctly | Pass   |

## 9. OUTPUT

Output 1: Styled Technology Hub webpage with navigation, cards, hover effects, buttons, and footer.



## 10. RESULT

Thus, the styled Technology Hub webpage was successfully designed and implemented using HTML5 and CSS3. The webpage was verified for navigation styling, Flexbox layout, card design, hover effects, buttons, and footer presentation.